

Algebraic Topology: a beginner's course

This lecture:

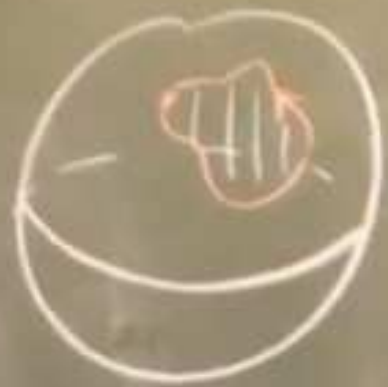
**AlgTop5: Two-dimensional
objects--the torus and genus**

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Lecture 5: Two-dimensional objects - the and genus



Ex



S^2

any loop disconnects S^2
Jordan curve thm

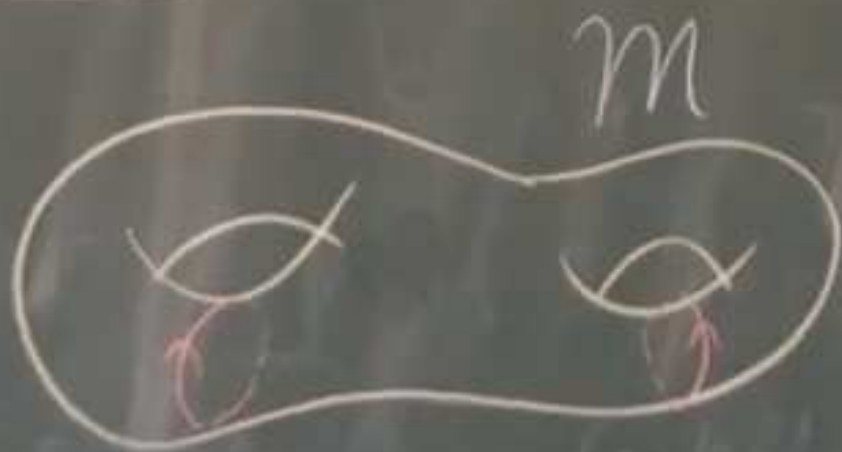
$$g(S^2) = 0$$

Ex



$$g(T) = 1$$

Ex



M

$$g(M) = 2$$

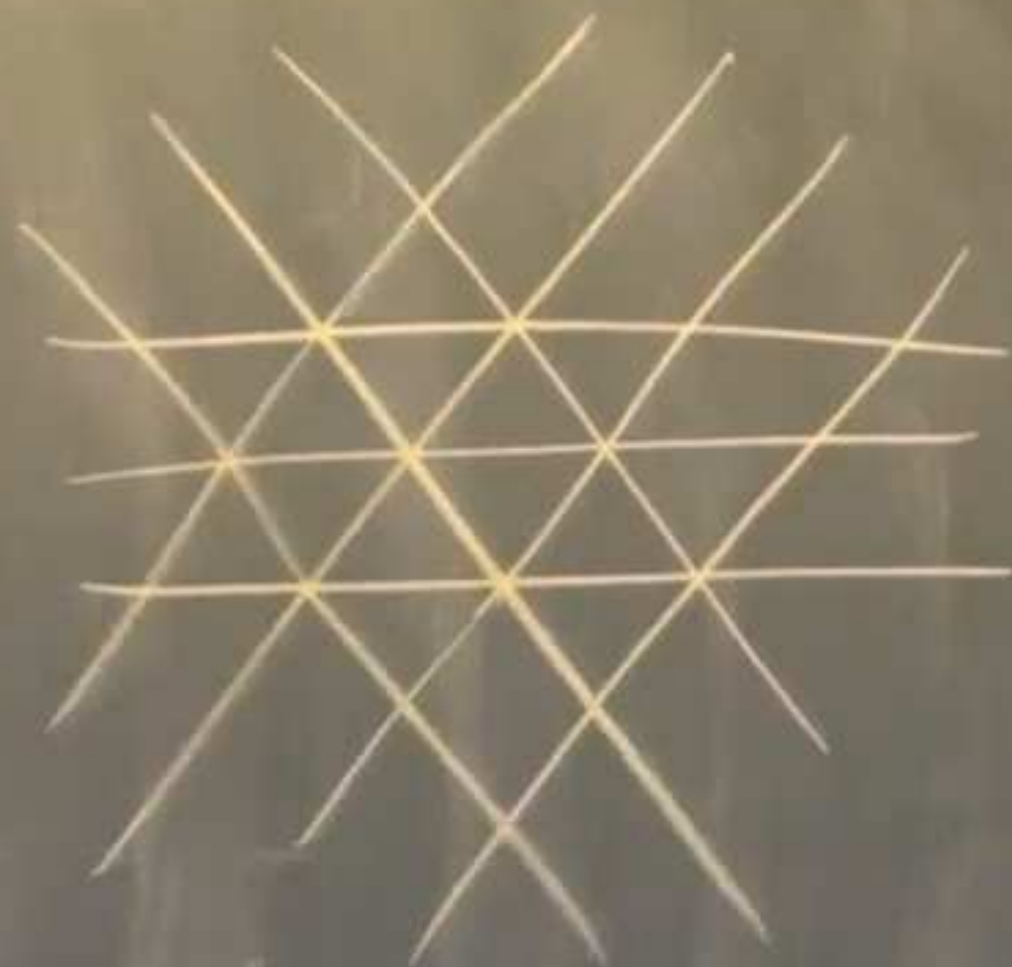
holes"

number of
holes on a surface
disconnect it.

tion of plane (regular)



squares



triangles



hexagons

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next in the series:

**AlgTop6: Non-orientable
surfaces--the Mobius band**